

improving things over time. The improvements that have happened are on multiple fronts. On the creation side, there have been significant improvements in the tools. As we've engaged with new artists and content creators, and as we've got feedback from them, we've constantly improved the technology in terms of creation. Also, what has happened due to the pandemic-induced lockdown is that a lot of studios were not accessible. People were mixing music at home, so there were a lot of mixes created on headphones. That meant there were improvements on the headphone mixing side. On the playback side, the technology and software that goes into the devices have also been constantly improved and refined. There's something called a virtualiser and renderer that takes an Atmos song and then renders this experience around the head. There have been a lot of improvements on this side as well to make sure there's a more accurate translation from creation to playback. So, what you're hearing is exactly the way the creator has created it.

Q What are some of the tools and devices that a user will need to experience Dolby Atmos Music?

Mr Pankaj Kedia: There are predominantly four listening use cases we'd think about from a music perspective. One is the mobile experience which could be from headphones or onboard speakers. Interestingly, we've seen a lot of data that indicates that listening from onboard speakers is much more common than one would imagine. Some services reported that up to 50 per cent of the listening is through onboard speakers. It's an important use case. In the case of mobile, you need an Atmos-compatible mobile phone which is very easy to find. Next, you can use any set of headphones with that, so it's not headphone dependant. Each headphone performs differently since they have their own frequency response and tonal profile. However, for an immersive experience, any headphones will give you that. If it's an Atmos compatible phone, it will play Atmos

Music from the speaker. You can hold the phone in front of you and it will give you the sense that sound is coming from behind you and all around you. Onboard speakers are pretty small, so what we can achieve with these speakers is pretty incredible.

Secondly, for the PC experience, you need one which is Dolby Atmos compatible. You can again listen to it on onboard speakers or through headphones. For example, Lenovo has a lot of models with Dolby Atmos capability, and there are others as well. Thirdly, is the living room experience. In the living room, you can be watching it from a connected set top box or smart TV. All the streaming apps now have living room apps as well. For example, if you had Apple Music on your Apple TV and you had an Atmos soundbar connected to it, you can then get the full experience in your living room. The fourth is something we're working on, it's a new area for us, it is on the automobile side. A lot of music listening happens in automobiles. From the demos we've done, people cannot believe how good and different it sounds.

Q Speaking of Apple, the company recently announced spatial audio on Apple Music with Dolby Atmos for Music. Can you tell us more about this partnership and what users can expect?

Mr Pankaj Kedia: The Apple partnership is a very special partnership for us, we've worked very closely with Apple for a long time. Apple has used our technology, both on the audio side with Atmos and also on the video side with Dolby Vision for a long time across all of their devices. As far as this announcement is concerned, Apple Music is one of the leading global music streaming services. They have made a big move in terms of redefining the music listening experience for their consumers. The music experience, for a long time, has remained unchanged in a certain way. We had the era of CDs, with MP3 it actually went down, it became worse because you were getting low-quality music where most of what was recorded were lost. With all of this, issues of pi-

racy also came around. It's now coming back where streaming services are offering high-quality content. Apple now takes it to another level where they are offering an immersive experience through spatial audio which is powered by Dolby Atmos Music. So, all of the spatial audio experience as far as Apple Music is concerned is the Dolby Atmos experience. Those artists and labels that have created music in Dolby Atmos will playback as a spatial music experience on Apple.

Q What are the things people will experience from Dolby Atmos Music in India? Are there going to be more platforms and services? What can the Indian audience expect?

Mr Pankaj Kedia: From an India perspective, we have Apple Music now, we have other services, and in the future, we may have even more services. What is most important for our team from an Indian perspective is getting Atmos to be used by all the artists across languages. Whether it's vernacular languages, or Hindi, any language, we want to make sure that artists have access to be able to create music in Atmos and then have platforms where they can publish their music with Atmos capabilities. The focus for us at this stage is around expanding the content availability. We've already launched a large number of Atmos titles with leading artists such as A. R. Rahman, Amit Trivedi, Divine, and Badshah. New tracks are being created every day and every week. So, the focus is to expand that and expand it across every language in India. Music is global but very local also, so we want to make sure whatever music you listen to, you should have to option to listen to it in Atmos.

Q Let's end with the future plans from Dolby Atmos and what users can expect next?

Mr Pankaj Kedia: We're doing a lot of interesting work on the audio-video side with television and OTT. Our DNA is around innovation so users will see a lot of interesting things coming out from us in the future. **Q**